# Bridge Platform - PostgreSQL Database Documentation

## SmartHire: Predictive Freelance Management Platform

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## Overview

This document describes the PostgreSQL database schema for the Bridge Platform, a predictive freelance management platform connecting freelancers with startups/companies. The database is designed with:

- **26 tables** organized into 11 logical modules
- **UUID primary keys** for global uniqueness and security
- **Proper normalization** (3NF) with strategic denormalization for performance
- **Comprehensive foreign key constraints** for data integrity
- **JSONB fields** for flexible metadata storage
- **Array types** for PostgreSQL-specific optimizations
- **Complete Admin Management System** for platform oversight

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# Database Architecture

## Module Structure

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┌──────────────────────────────────────────────────────────────────────┐
│                        BRIDGE DATABASE SCHEMA                        │
├──────────────────────────────────────────────────────────────────────┤
│  1. Identity & User Management (5 tables)                            │
│     ├── roles                                                        │
│     ├── users                                                        │
│     ├── freelancers                                                  │
│     ├── companies                                                    │
│     └── admins                                                       │
├──────────────────────────────────────────────────────────────────────┤
│  2. Skills Management (3 tables)                                     │
│     ├── skills                                                       │
│     ├── freelancer_skills                                            │
│     └── mission_skills                                               │
├──────────────────────────────────────────────────────────────────────┤
│  3. Mission & Project Management (3 tables)                          │
│     ├── missions                                                     │
│     ├── applications                                                 │
│     └── mission_skills                                               │
├──────────────────────────────────────────────────────────────────────┤
│  4. Contract & Deliverables (3 tables)                               │
│     ├── contracts                                                    │
│     ├── milestones                                                   │
│     └── deliverables                                                 │
├──────────────────────────────────────────────────────────────────────┤
│  5. Payment & Escrow (3 tables)                                      │
│     ├── wallets                                                      │
│     ├── transactions                                                 │
│     └── escrow                                                       │
├──────────────────────────────────────────────────────────────────────┤
│  6. Reviews & Ratings (1 table)                                      │
│     └── reviews                                                      │
├──────────────────────────────────────────────────────────────────────┤
│  7. Time Tracking & Activity (2 tables)                              │
│     ├── time_entries                                                 │
│     └── activity_logs                                                │
├──────────────────────────────────────────────────────────────────────┤
│  8. Communication (3 tables)                                         │
│     ├── conversations                                                │
│     ├── conversation_participants                                    │
│     └── messages                                                     │
├──────────────────────────────────────────────────────────────────────┤
│  9. Notifications (1 table)                                          │
│     └── notifications                                                │
├──────────────────────────────────────────────────────────────────────┤
│  10. AI & Analytics (2 tables)                                       │
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│      ├── ai_match_scores                                                  │
│      └── freelancer_analytics                                             │
├──────────────────────────────────────────────────────────────────────────┤
│  11. Admin Management (4 tables)                                          │
│      ├── admin_actions                                                    │
│      ├── platform_settings                                                │
│      ├── support_tickets                                                  │
│      └── ticket_messages                                                  │
└──────────────────────────────────────────────────────────────────────────┘
```

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## Table Definitions

### 1. **IDENTITY & USER MANAGEMENT**

`roles`

Lookup table for user role definitions.

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | UUID | PK | Unique role identifier |
| name | VARCHAR(50) | NOT NULL, UNIQUE | Role name (admin, freelancer, company, guest) |
| description | TEXT | | Role description |
| permissions | JSONB | | Role permissions as JSON |
| created_at | TIMESTAMP | DEFAULT now() | Creation timestamp |

`users`

Central authentication table - all users regardless of role.

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | UUID | PK | Unique user identifier |
| email | VARCHAR(255) | NOT NULL, UNIQUE | User email address |
| password_hash | VARCHAR(255) | NOT NULL | Bcrypt hashed |

| Column | Type | Constraints | Description |
|---|---|---|---|
| | | | password |
| role_id | UUID | FK → roles.id | User's role |
| is_verified | BOOLEAN | DEFAULT false | Email verification status |
| is_active | BOOLEAN | DEFAULT true | Account active status |
| is_suspended | BOOLEAN | DEFAULT false | Account suspended |
| suspended_by | UUID | FK → admins.id | Admin who suspended |
| suspension_reason | TEXT | | Reason for suspension |
| suspended_at | TIMESTAMP | | Suspension timestamp |
| last_login | TIMESTAMP | | Last login timestamp |
| created_at | TIMESTAMP | DEFAULT now() | Account creation |
| updated_at | TIMESTAMP | DEFAULT now() | Last update |

`freelancers`

Freelancer profile extension - 1:1 relationship with users.

| Column | Type | Constraints | Description |
|---|---|---|---|
| user_id | UUID | FK → users.id, UQ | Link to auth user |
| first_name | VARCHAR(100) | NOT NULL | First name |
| last_name | VARCHAR(100) | NOT NULL | Last name |
| headline | VARCHAR(255) | | Professional headline |
| bio | TEXT | | Detailed bio |

| Column | Type | Constraints | Description |
|---|---|---|---|
| hourly_rate | DECIMAL (10,2) | | Default hourly rate |
| availability_status | VARCHAR(50) | DEFAULT ‘available’ | availability, busy, etc. |
| location | VARCHAR(255) | | Physical location |
| timezone | VARCHAR(50) | | User timezone |
| languages | TEXT[] | | Spoken languages |
| portfolio_url | VARCHAR(255) | | Portfolio website |
| linkedin_url | VARCHAR(255) | | LinkedIn profile |
| github_url | VARCHAR(255) | | GitHub profile |
| ai_aggregate_score | DECIMAL (3,2) | DEFAULT 0.00 | AI-calculated score (0-5) |
| total_earnings | DECIMAL (15,2) | DEFAULT 0.00 | Lifetime earnings |
| completed_missions_count | INTEGER | DEFAULT 0 | Completed missions |
| success_rate | DECIMAL (5,2) | DEFAULT 0.00 | Success percentage |
| is_profile_public | BOOLEAN | DEFAULT true | Profile visibility |
| created_at | TIMESTAMP | DEFAULT now() | Profile creation |

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| Column | Type | Constraints | Description |
|---|---|---|---|
| updated_at | TIMESTAMP | DEFAULT now() | Last update |

`Startups`

Startup profile extension - 1:1 relationship with users.

| Column | Type | Constraints | Description |
|---|---|---|---|
| user_id | UUID | FK → users.id, UQ/PK | Link to auth user |
| startup_name | VARCHAR(255) | NOT NULL | Company name |
| description | TEXT | | Startups description description |
| industry | VARCHAR(100) | | Industry sector |
| startup_size | VARCHAR(50) | | Company size range |
| website_url | VARCHAR(255) | | Company website |
| location | VARCHAR(255) | | Company location |
| logo_url | VARCHAR(255) | | Company logo |
| verification_status | VARCHAR(50) | DEFAULT ‘pending’ | Verification status |
| total_missions_posted | INTEGER | DEFAULT 0 | Total missions posted |
| total_spent | DECIMAL(15,2) | DEFAULT 0.00 | Total amount spent |
| average_ratin | DECIMAL | DEFAULT 0.00 | Average rating received |

| Column | Type | Constraints | Description |
|---|---|---|---|
| g | (3,2) | | |
| created_at | TIMESTAMP | DEFAULT now() | Profile creation |
| updated_at | TIMESTAMP | DEFAULT now() | Last update |

`admins`

Administrator profile extension - 1:1 relationship with users.

| Column | Type | Constraints | Description |
|---|---|---|---|
| user_id | UUID | FK → users.id, UQ | Link to auth user |
| first_name | VARCHAR(100) | NOT NULL | First name |
| last_name | VARCHAR(100) | NOT NULL | Last name |
| admin_level | VARCHAR(50) | DEFAULT ‘standard’ | super, standard, support |
| department | VARCHAR(100) | | Admin department |
| permissions | JSONB | | Custom permissions |
| can_manage_users | BOOLEAN | DEFAULT true | User management permission |
| can_manage_missions | BOOLEAN | DEFAULT true | Mission management permission |
| can_manage_payments | BOOLEAN | DEFAULT false | Payment management permission |
| can_view_analytics | BOOLEAN | DEFAULT true | Analytics access |
| can_imperson | BOOLEA | DEFAULT false | User impersonation |

| Column | Type | Constraints | Description |
|---|---|---|---|
| ate | N | | permission |
| last_active_at | TIMESTAMP | | Last activity |
| created_at | TIMESTAMP | DEFAULT now() | Profile creation |
| updated_at | TIMESTAMP | DEFAULT now() | Last update |

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## 2. SKILLS MANAGEMENT

`Skills(user services)`

Master skills catalog.

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | UUID | PK | Unique skill ID |
| name | VARCHAR(100) | NOT NULL, UNIQUE | Skill name |
| category | VARCHAR(100) | | Skill category |
| description | TEXT | | Skill description |
| created_by | UUID | FK → admins.id | Admin who created |
| is_active | BOOLEAN | DEFAULT true | Active status |
| created_at | TIMESTAMP | DEFAULT now() | Creation timestamp |

`freelancer_skills(user services)`

Junction table: freelancers ↔ skills.

| Column | Type | Constraints | Description |
|---|---|---|---|
| freelancer_id | UUID | FK → freelancers.id | Freelancer reference |
| skill_id | UUID | FK → skills.id | Skill reference |
| proficiency_level | INTEGER | | Skill level (1-5) |
| years_of_experience | INTEGER | | Years of experience |
| is_primary | BOOLEAN | DEFAULT false | Primary skill flag |

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| Column | Type | Constraints | Description |
|---|---|---|---|
| created_at | TIMESTAMP | DEFAULT now() | Creation timestamp |

**Unique Constraint:** (freelancer_id, skill_id)

`mission_skills`(`mission microservices`)

Junction table: missions ↔ required skills.

| Column | Type | Constraints | Description |
|---|---|---|---|
| mission_id | UUID | FK->missions.id | Mission |
| skill_id | UUID | FK-> skills.id | Skill |
| is_required | BOOLEAN | DEFAULT true | Required vs preferred |
| proficiency_min | INTEGER | | Minimum skill level (1-5) |
| created_at | TIMESTAMP | DEFAULT now() | Creation timestamp |

**Unique Constraint:** (mission_id, skill_id)

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## 3. MISSION & PROJECT MANAGEMENT

`Missions`(`mission service`)

Job postings created by companies.

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | UUID | PK | Unique mission ID |
| Startups_id | UUID | Startups.id | startup(Api to user services) |
| title | VARCHAR(255) | NOT NULL | Mission title |
| description | TEXT | NOT NULL | Mission description |
| requirements | TEXT | | Detailed requirements |
| budget_min | DECIMAL (12,2) | | Minimum budget |
| budget_max | DECIMAL (12,2) | | Maximum budget |

| Column | Type | Constraints | Description |
|---|---|---|---|
| budget_type | VARCHAR(50) | DEFAULT ‘fixed’ | fixed, hourly, range |
| duration_days | INTEGER | | Estimated duration |
| experience_level | VARCHAR(50) | | Required experience |
| location_type | VARCHAR(50) | DEFAULT ‘remote’ | remote, onsite, hybrid |
| status | VARCHAR(50) | DEFAULT ‘draft’ | draft, open, in_progress, completed, cancelled, closed |
| visibility | VARCHAR(50) | DEFAULT ‘public’ | public, private, invite_only |
| application_deadline | TIMESTAMP | | Application deadline |
| ai_match_enabled | BOOLEAN | DEFAULT true | AI matching enabled |
| views_count | INTEGER | DEFAULT 0 | View counter |
| applications_count | INTEGER | DEFAULT 0 | Application counter |
| is_flagged | BOOLEAN | DEFAULT false | Flagged for review |
| flagged_by | UUID | FK → admins.id | Admin who flagged |
| flagged_reason | TEXT | | Flag reason |
| flagged_at | TIMESTAMP | | Flag timestamp |
| reviewed_by | UUID | FK → admins.id | Admin who reviewed |

| Column | Type | Constraints | Description |
|---|---|---|---|
| review_notes | TEXT | | Admin review notes |
| created_at | TIMESTAMP | DEFAULT now() | Creation timestamp |
| updated_at | TIMESTAMP | DEFAULT now() | Last update |
| published_at | TIMESTAMP | | Publication timestamp |

`applications`

Freelancer applications to missions.(mission service)

| Column | Type | Constraints | Description |
|---|---|---|---|
| mission_id | UUID | FK →missions.id | Target mission |
| freelancer_id | UUID | FK->freelancers.id | Applicant |
| cover_letter | TEXT | | Application message |
| proposed_rate | DECIMAL(12,2) | | Proposed rate |
| proposed_duration_days | INTEGER | | Proposed duration |
| status | VARCHAR(50) | DEFAULT ‘pending’ | pending, viewed, shortlisted, accepted, rejected, withdrawn |
| ai_match_score | DECIMAL(5,2) | | AI match score (0-100) |
| company_notes | TEXT | | Internal company notes |
| viewed_at | TIMESTAMP | | When viewed by company |
| responded_at | TIMESTA | | When response sent |

| Column | Type | Constraints | Description |
|---|---|---|---|
| | MP | | |
| created_at | TIMESTAMP | DEFAULT now() | Application timestamp |
| updated_at | TIMESTAMP | DEFAULT now() | Last update |

**Unique Constraint:** (mission_id, freelancer_id) - One application per freelancer per mission

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## 4. CONTRACT & DELIVERABLES

`Contracts(Contra microservice)`

Work agreements between freelancer and company.

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | UUID | PK | Unique contract ID |
| mission_id | UUID | FK_>missions.id, UQ | |
| freelancer_id | UUID | FK_>freelancers.id | |
| contract_terms | TEXT | | Contract terms |
| agreed_rate | DECIMAL(12,2) | NOT NULL | Agreed rate |
| agreed_duration_days | INTEGER | NOT NULL | Agreed duration |
| start_date | TIMESTAMP | | Contract start |
| end_date | TIMESTAMP | | Contract end |
| status | VARCHA | DEFAULT ‘pending’ | pending, active, completed, disputed, |

| Column | Type | Constraints | Description |
|---|---|---|---|
| | R(50) | | cancelled |
| freelancer_signed | BOOLEAN | DEFAULT false | Freelancer signed |
| company_signed | BOOLEAN | DEFAULT false | Company signed |
| freelancer_signed_at | TIMESTAMP | | Freelancer signature time |
| company_signed_at | TIMESTAMP | | Company signature time |
| completed_at | TIMESTAMP | | Completion timestamp |
| cancellation_reason | TEXT | | Cancellation reason |
| is_disputed | BOOLEAN | DEFAULT false | Dispute flag |
| dispute_reason | TEXT | | Dispute reason |
| disputed_at | TIMESTAMP | | Dispute timestamp |
| resolved_by | UUID | FK → admins.id | Admin who resolved |
| resolution_notes | TEXT | | Resolution notes |
| resolved_at | TIMESTAMP | | Resolution timestamp |
| created_at | TIMESTAMP | DEFAULT now() | Creation timestamp |
| updated_at | TIMESTA | DEFAULT now() | Last update |

| Column | Type | Constraints | Description |
|---|---|---|---|
| | MP | | |

`Milestones(contract)`

Project milestones within contracts.

| Column | Type | Constraints | Description |
|---|---|---|---|
| id | UUID | PK | Unique milestone ID |
| contract_id | UUID | FK → contracts.id | Parent contract |
| title | VARCHAR(255) | NOT NULL | Milestone title |
| description | TEXT | | Milestone description |
| due_date | TIMESTAMP | | Due date |
| amount | DECIMAL(12,2) | | Payment amount |
| status | VARCHAR(50) | DEFAULT ‘pending’ | pending, in_progress, completed, approved |
| completed_at | TIMESTAMP | | Completion timestamp |
| approved_at | TIMESTAMP | | Approval timestamp |
| order_index | INTEGER | DEFAULT 0 | Display order |
| created_at | TIMESTAMP | DEFAULT now() | Creation timestamp |

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## 10. AI & ANALYTICS

`ai_match_scores(Analyse micro services)`

AI-calculated match scores between freelancers and missions.

| Column | Type | Constraints | Description |
|---|---|---|---|

| Column | Type | Constraints | Description |
|---|---|---|---|
| mission_id | UUID | FK->missions.id | |
| freelancer_id | UUID | FK->freelancers.id | |
| overall_score | DECIMAL(5,2) | NOT NULL | Overall match (0-100) |
| skills_match_score | DECIMAL(5,2) | | Skills match |
| experience_match_score | DECIMAL(5,2) | | Experience match |
| rate_match_score | DECIMAL(5,2) | | Rate compatibility |
| availability_match_score | DECIMAL(5,2) | | Availability match |
| past_performance_score | DECIMAL(5,2) | | Performance factor |
| matching_reasons | JSONB | | Explanation data |
| calculated_at | TIMESTAMP | DEFAULT now() | Calculation timestamp |

**Unique Constraint:** (mission_id, freelancer_id)

`freelancer_analytics`(`Analyse microservice`)

Aggregated analytics for freelancers.

| Column | Type | Constraints | Description |
|---|---|---|---|
| freelancer_id | UUID | FK->freelancers.id, | |
| response_time_avg_minutes | INTEGER | | Average response time |
| on_time_delivery_rate | DECIMAL(5,2) | | On-time delivery % |
| client_satisfaction_rate | DECIMAL(5,2) | | Client satisfaction % |
| repeat_client_rate | DECIMAL(5,2) | | Repeat client % |
| communication_score | DECIMAL(5,2) | | Communication score |

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| Column | Type | Constraints | Description |
|---|---|---|---|
| reliability_score | DECIMAL(5,2) | | Reliability score |
| stress_indicator | DECIMAL(5,2) | | Stress level indicator |
| productivity_score | DECIMAL(5,2) | | Productivity score |
| calculated_at | TIMESTAMP | DEFAULT now() | Calculation timestamp |

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